

## Homework Exercise Sheet

## Ex. 9 Solution

Label (S) : only one correct answer; (M) : more than one correct answer

1. (M) Which of the following statements regarding fixed-point iterations  $x^{(k+1)} = \Phi(x^{(k)})$  for solving the non-linear equation  $F(x^*) = 0$  is **correct**?
- a) If  $\Phi$  is consistent, then any convergent sequence  $(x^{(k)})_k$  converges to  $x^*$ .
  - b) If  $\Phi$  is consistent and contractive, then it converges globally.
  - c) If  $\Phi$  is consistent and contractive, then, if it converges, it converges (at least) linearly.
  - d) If  $\Phi$  is consistent and  $\|D\Phi(x^*)\| < 1$ , then it converges locally.
  - e) If  $\Phi$  is consistent and  $\|D\Phi(x^*)\| < 1$ , then, if it converges, it converges (at least) linearly.

**Hint:** Recall that

- $\Phi : \mathcal{D} \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  is **consistent** with  $F$ , if  $F(x^*) = 0 \Leftrightarrow x^* = \Phi(x^*)$ .
- $\Phi : \mathcal{D} \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  is **contractive** (w.r.t. norm  $\|\cdot\|$ ), if  $\|\Phi(x) - \Phi(y)\| \leq L\|x - y\|$  with  $L < 1$  for all  $x, y \in \mathcal{D}$ .
- $\Phi : \mathcal{D} \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  **converges locally**, if there exists a neighborhood  $\mathcal{U} \subset \mathcal{D}$  such that  $x^{(0)} \in \mathcal{U} \Rightarrow \lim_{k \rightarrow \infty} x^{(k)} = x^*$ . If  $\mathcal{U} = \mathcal{D}$ ,  $\Phi$  is called to be **converging globally**.

**Solution:**

**a), b), c), d), e).** Statement a) is shown in Theorem 8.2.1.8. Statement b) and c) are shown in Theorem 8.3.2.6. Statements e) and f) are shown in Theorem 8.3.2.7.

2. (S) We use the bisection method to find a zero of  $f : [-100, 100] \rightarrow \mathbb{R}$  with  $f(1) < 0$  and  $f(2) > 0$ . Find an a priori bound for the number of steps needed to determine a zero with a guaranteed relative error of  $10^{-6}$ .

a)  $\approx 10$                       b)  $\approx 20$                       c)  $\approx 30$                       d)  $\approx 10^6$                       e)  $\approx 10^5$

**Solution:**

**b).** By discussion in 8.4.1.1, the number of steps needed is approximately  $\log_2 \left( \frac{|b-a|}{\text{tol}} \right)$ . In this case, it is  $\log_2 \left( \frac{|2-1|}{10^{-6}} \right) \approx 20$ .

3. (M) Given a nonlinear equation  $F(x) = x, x \in \mathbb{R}$  that admits a unique solution  $x^*$ , please select **consistent** fixed-point iteration functions  $\Phi : \mathbb{R} \rightarrow \mathbb{R}$ .

- a)  $\Phi(x) := F(x)$
- b)  $\Phi(x) := F(x) - x$
- c)  $\Phi(x) := F^2(x) - F(x) + 2x$
- d)  $\Phi(x) := -F^2(x) + 2xF(x) + F(x) - x^2$
- e)  $\Phi(x) := \frac{F^2(x) - xF(x) + 1}{F(x) - x}$
- f)  $\Phi(x) := \frac{x F'(x) - F(x)}{F'(x) - 1}$

**Solution:**

**a), f)** It is straightforward to check that a), d), f) satisfy that  $\Phi(x^*) = x^*$ . However, one should notice that d) admits another fixed-point  $x^{**}$  such that  $F(x^{**}) = x^{**} + 1$ .

4. **(M)** Consider the iteration  $\mathbf{x}^{(k+1)} = \mathbf{A}\mathbf{x}^{(k)} \in \mathbb{R}^2$  where

$$\mathbf{A} = \begin{bmatrix} 1/3 & 1 \\ 0 & 1/2 \end{bmatrix}.$$

Select correct statements.

- a)  $(\mathbf{x}^{(k)})$  converges for any initial guess  $\mathbf{x}^{(0)} \in \mathbb{R}^2$ .
- b)  $(\mathbf{x}^{(k)})$  converges only for certain initial guess  $\mathbf{x}^{(0)} \in \mathbb{R}^2$ .
- c) If  $(\mathbf{x}^{(k)})$  converges,  $\lim_{k \rightarrow \infty} \mathbf{x}^{(k)} = [0, 0]^\top$ .
- d) If  $(\mathbf{x}^{(k)})$  converges,  $\lim_{k \rightarrow \infty} \mathbf{x}^{(k)} = [1/3, 1/2]^\top$ .
- e)  $(\mathbf{x}^{(k)})$  converges linearly for any initial guess  $\mathbf{x}^{(0)} \in \mathbb{R}^2$ .
- f)  $(\mathbf{x}^{(k)})$  converges linearly only for certain initial guess  $\mathbf{x}^{(0)} \in \mathbb{R}^2$ .

**Hint:** The matrix  $A$  can be diagonalized.

**Solution:**

**a), c), f).** The diagonalization of  $A$  is  $A = PDP^{-1}$ , where  $D = \text{diag}(1/3, 1/2)$  is a diagonal matrix of eigenvalues of  $A$ . Then, by introducing  $y^{(k)} = P^{-1}x^{(k)}$  and making a set of rewrites:  $x^{(k+1)} = Ax^{(k)}$ ,  $x^{(k+1)} = PDP^{-1}x^{(k)}$ ,  $P^{-1}x^{(k+1)} = DP^{-1}x^{(k)}$ ,  $y^{(k+1)} = Dy^{(k)}$ ; the limit can be found independently for each component of the  $y^{(k)}$  vector:  $\lim_{k \rightarrow \infty} y^{(k)} = [0, 0]^\top$ . Finding the exact value of matrix  $P$  is not required (although possible by solving  $AP = PD$ ) to see that  $x^* = \lim_{k \rightarrow \infty} x^{(k)} = \lim_{k \rightarrow \infty} Py^{(k)} = 0$ . The fact of linear convergence can be disproved by observing that for  $x^{(k)} = [0, 2]^\top$ ,  $x^{(k+1)} = [2, 1]^\top$ . Therefore, there is no  $L < 1$  for which  $\|x^{(k+1)} - x^*\| \leq L\|x^{(k)} - x^*\|$ .

5. **(M)** Given  $a > 0$  the following iteration functions spawn fixed-point iterations for the computation of  $\sqrt{a}$ :

- $\varphi_1(x) = a + x - x^2$ ,
- $\varphi_2(x) = a/x$ ,
- $\varphi_3(x) = 1 + x - x^2/a$ ,
- $\varphi_4(x) = (x + a/x)/2$ .

Predict the behavior and the type of convergence of the induced fixed-point iterations when started with an initial guess *sufficiently close* to  $\sqrt{a}$ .

- |                                                              |                                                              |
|--------------------------------------------------------------|--------------------------------------------------------------|
| a) $\varphi_1$ does not converge.                            | b) $\varphi_1$ converges for <i>sufficiently small</i> $a$ . |
| c) $\varphi_1$ converges for <i>sufficiently large</i> $a$ . | d) $\varphi_1$ converges quadratically.                      |
| e) $\varphi_2$ does not converge.                            | f) $\varphi_2$ converges for <i>sufficiently small</i> $a$ . |
| g) $\varphi_2$ converges for <i>sufficiently large</i> $a$ . | h) $\varphi_2$ converges quadratically.                      |
| i) $\varphi_3$ does not converge.                            | j) $\varphi_3$ converges for <i>sufficiently small</i> $a$ . |
| k) $\varphi_3$ converges for <i>sufficiently large</i> $a$ . | l) $\varphi_3$ converges quadratically.                      |
| m) $\varphi_4$ does not converge.                            | n) $\varphi_4$ converges for <i>sufficiently small</i> $a$ . |
| o) $\varphi_4$ converges for <i>sufficiently large</i> $a$ . | p) $\varphi_4$ converges quadratically.                      |

**Solution:**

**b), e), k), n), o), p).**  $\varphi'_1(x) = 1 - 2x$ ,  $\varphi'_2(x) = -a/x^2$ ,  $\varphi'_3(x) = 1 - 2x/a$ ,  $\varphi'_4(x) = (1 - a/x^2)/2$ .  
 $\varphi'_1(\sqrt{a}) = 1 - 2\sqrt{a}$ , hence, it linearly converges for sufficiently small  $a$  when  $|\varphi'_1(\sqrt{a})| < 1$ .  
 $\varphi'_2(\sqrt{a}) = -1$ , hence, it does not converge  $|\varphi'_2(\sqrt{a})| \geq 1$ .  $\varphi'_3(\sqrt{a}) = 1 - 2/\sqrt{a}$ , hence, it linearly converges for sufficiently large  $a$  when  $|\varphi'_3(\sqrt{a})| < 1$ .  $\varphi'_4(\sqrt{a}) = 0$  and  $\varphi''_4(\sqrt{a}) = 1/\sqrt{a}$ , hence, it converges quadratically for any  $a$ .

6. **(S)** To check if the following fixed-point iteration converges, **compute the norm of the Jacobian matrix** induced by the  $L^1$  vector norm at the point  $x^* = 1.8$ ,  $y^* = 0.4$ ,  $z^* = 2.7$ . Compute approximations rounded to one decimal place.

$$\begin{aligned}\Phi_1(x, y, z) &= \sqrt{0.5(7 - (y - 1)^2)} \\ \Phi_2(x, y, z) &= 3 - e^{xy} - \sin z \\ \Phi_3(x, y, z) &= 5 - 3xy\end{aligned}$$

- |         |        |         |        |
|---------|--------|---------|--------|
| a) -6.6 | b) 3.3 | c) -8.8 | d) 5.5 |
|---------|--------|---------|--------|

**Hint:** Use the following approximations  $e^{x^*y^*} = 2.0$ ,  $\cos z^* = -0.9$ ,  $\sqrt{13.3} = 3.6$ .

**Solution:**

**c).** First, find the Jacobian matrix  $\begin{bmatrix} \frac{\partial \Phi_1}{\partial x} & \frac{\partial \Phi_1}{\partial y} & \frac{\partial \Phi_1}{\partial z} \\ \frac{\partial \Phi_2}{\partial x} & \frac{\partial \Phi_2}{\partial y} & \frac{\partial \Phi_2}{\partial z} \\ \frac{\partial \Phi_3}{\partial x} & \frac{\partial \Phi_3}{\partial y} & \frac{\partial \Phi_3}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1-y}{\sqrt{-2y^2+4y+12}} & 0 \\ -ye^{xy} & -xe^{xy} & -\cos z \\ -3y & -3x & 0 \end{bmatrix}$ . Then

evaluate it at the point  $(x^*, y^*, z^*)$ :  $\begin{bmatrix} 0 & 0.2 & 0 \\ -0.8 & -3.6 & 0.9 \\ -1.2 & -5.4 & 0 \end{bmatrix}$ . Next, compute the matrix norm induced by the  $L^1$  vector norm by finding the maximum over per-column sums of absolute values.

7. (S) What is the convergence rate of following iterative formula for calculating  $\pi$ :

$$\begin{aligned} a_0 &= 0, g_0 = 1 \\ a_n &= \frac{a_{n-1} + g_{n-1}}{2}, g_n = \sqrt{a_n g_{n-1}} \\ \hat{\pi}_n &= \frac{2}{a_n} \end{aligned}$$

**Hint:** Check Remark 8.2.2.12 for calculation of convergence rate.

a) 0.5

b) 1

c) 1.5

d) 2

**Solution:**

**b).** According to Remark 8.2.2.12, we can estimate the convergence rate by

$$p \approx \frac{\log \epsilon_{n+1} - \log \epsilon_n}{\log \epsilon_n - \log \epsilon_{n-1}}, \epsilon_n = |\hat{\pi}_n - \pi|.$$

Now we can tabulate  $\epsilon_n$  to calculate the convergence rate:

| $n$ | $\epsilon_n$       | $p$          |
|-----|--------------------|--------------|
| 1   | 0.8584073464       |              |
| 2   | 0.1721158454       | 0.8926868401 |
| 3   | 0.04100522448      | 0.974560201  |
| 4   | 0.01013225384      | 0.9937175586 |
| 5   | 0.002525731656     | 0.9984341118 |
| 6   | 0.0006309763527    | 0.9996088213 |
| 7   | 0.0001577155792    | 0.9999022236 |
| 8   | 0.00003942711339   | 0.999975557  |
| 9   | 0.000009856667016  | 0.9999938893 |
| 10  | 0.000002464159796  | 0.9999984723 |
| 11  | 0.0000006160395141 | 0.9999996181 |
| 12  | 0.0000001540098513 | 0.9999999045 |
| 13  | 3.850246114E-08    | 0.9999999761 |
| 14  | 9.625615178E-09    | 0.999999994  |
| 15  | 2.406403788E-09    | 0.9999999985 |
| 16  | 6.016009465E-10    | 0.9999999996 |
| 17  | 1.504002366E-10    | 0.9999999999 |
| 18  | 3.760005915E-11    | 1            |
| 19  | 9.400014788E-12    | 1            |

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